DEEP LEARNING COURSE PROJECT • SEMESTER 5

ROAD HEALTH AI

Automated AI Pavement Distress Assessment, Intelligent Defect Scoring, and Municipal Infrastructure Governance Platform

A comprehensive platform report detailing system architecture, machine learning models, mathematical evaluation formulas, relational database structure, municipal contractor workflows, and all implemented modules without code listings.

PROJECT DOMAIN

Computer Vision • Municipal Infrastructure

PRIMARY AI MODELS

Dual YOLOv8 Ensemble (RDD2022 Benchmark)

CORE WEB FRAMEWORK

FastAPI (Python) • React (Vite SPA)

DATABASE STORAGE

SQLite Relational Database with SQLAlchemy

NOTIFICATION CHANNEL

Asynchronous TLS SMTP Email Dispatcher

TARGET AUDIENCE

Project Reviewers, Municipal Engineers, Contractors

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1. Executive Summary & Problem Formulation

1.1 Project Purpose & Real-World Motivation

Road infrastructure is the lifeblood of national transport, economic logistics, and public safety. Pavement surfaces endure continuous deterioration driven by heavy axle loads, monsoon rainwater seepage, sub-base compaction failure, and thermal contraction. Traditional municipal road inspection relies upon manual visual surveys conducted by civil engineering teams walking or driving along roadways.

These manual methodologies suffer from critical operational inefficiencies:

- **Subjectivity and Inconsistency:** Human evaluations vary widely across inspectors, lacking objective quantitative metrics.
- **Slow Detection Cycles:** Minor hairline cracks often remain undetected for months until they transform into massive vehicle-damaging potholes.
- **Disconnected Contractor Response:** Once defects are noted, assigning repairs to contractors responsible for specific geographical municipal wards involves manual paperwork and long administrative delays.
- **Citizen Disconnect:** Citizens reporting hazardous road conditions have no visibility into inspection results or scheduled repairs.

1.2 The Road Health AI Vision

Road Health AI provides an automated computer vision and governance platform that transforms standard road photographs into quantified structural ratings. By uploading a single road photo via mobile or desktop, the platform automatically detects distress locations, calculates a transparent 0 to 100 Road Health Score, maps the defect geographically, and dispatches actionable repair alerts directly to the designated contractor assigned to that exact city ward or neighborhood.

Key Platform Achievements:

- Dual-model deep learning vision pipeline detecting longitudinal cracks, transverse cracks, alligator cracking, and potholes.
- Out-of-domain asphalt validation classifier ensuring only real road pavement photographs are processed.
- Transparent mathematical formula balancing affected surface area and defect structural severity into a single health score.
- Multi-tenant architecture isolating citizen inspection history while giving contractors and administrators regional priority queues.
- Hierarchical geographical dispatching routing damage alerts directly to area-specific contractors and notifying administrators.

2. Comprehensive Technology Stack

The system is built on modern, decoupled architectural layers designed for high concurrency, low latency, and robust cross-platform accessibility.

Layer	Technology Selected	Purpose & Operational Role
Deep Learning Model	Ultralytics YOLOv8	State-of-the-art single-stage convolutional object detector performing real-time bounding box regression and classification on road distress.
ML Framework	PyTorch & Torchvision	Underlying deep learning tensor calculation engine executing GPU and CPU inference pipelines.
Computer Vision	OpenCV & Pillow	Image preprocessing, bounding box annotation, color-coded visualizer overlays, and surface feature analysis.
Backend Web API	FastAPI (Python 3.10)	Asynchronous high-performance web framework providing strict request validation, dependency injection, and OpenAPI specifications.
ASGI Web Server	Uvicorn	Production asynchronous server gateway interface running the FastAPI application with auto-reloading capability.
Relational ORM	SQLAlchemy	Object-Relational Mapping library translating Python model classes into database operations with relationship management.
Database Engine	SQLite	Embedded relational database engine providing ACID transaction safety, zero external server overhead, and high read throughput.
User Security	Passlib (BCrypt) & PyJWT	Cryptographic password hashing with random salt generation and stateless JSON Web Token generation for bearer authentication.
Alert Dispatcher	smtplib & email.message	Python standard library SMTP client communicating securely with Gmail SMTP servers via Transport Layer Security (TLS port 587).
Frontend Web App	React 18 & Vite	Single-page application framework delivering dynamic client-side rendering, hot module replacement, and declarative UI state.
Styling & Design	Tailwind CSS	Utility-driven responsive design system providing modern dark-mode themes, rounded card components, and fluid grids.
Interactive Mapping	Leaflet & React-Leaflet	Interactive JavaScript web map rendering geographic damage pins, coordinate markers, and full-screen location picker modals.
Reverse Geocoding	OpenStreetMap Nominatim	Public geospatial reverse geocoding web service translating latitude and longitude GPS coordinates into human-readable

Layer	Technology Selected	Purpose & Operational Role
		street addresses.
Data Visualization	Recharts	SVG charting library generating defect distribution bar charts, severity rings, and temporal health trends.

3. Machine Learning Models & Training Dataset

3.1 Training Dataset: Global Road Damage Dataset (RDD2022)

The core object detection model was trained and evaluated against the **RDD2022** benchmark dataset, compiled for the *Crowdsensing-based Road Damage Detection Challenge (CRDDC2022)*. RDD2022 encompasses more than 47,000 annotated road images captured across diverse international environments including **India, Japan, the Czech Republic, Norway, the United States, and China**.

The multi-national diversity of RDD2022 ensures that the model generalizes robustly across varying asphalt colorations, concrete pavements, shadow patterns, rainy reflections, and road markings.

3.2 Damage Class Taxonomy

The system classifies pavement distress into four standardized engineering categories:

Defect Code	Defect Name	Distress Characteristics	Severity Weight
D00	Longitudinal Crack	Cracks running parallel to the direction of vehicular travel. Often initiated by asphalt joint shrinkage, inadequate pavement width, or repeated wheel-load fatigue along wheel paths.	**1.0 ×**
D10	Transverse Crack	Cracks extending across the pavement perpendicular to the road centerline. Primarily caused by temperature variations, asphalt binder hardening, or structural reflection from underlying joints.	**1.5 ×**
D20	Alligator Crack	Series of interconnected multi-sided cracks resembling the skin of an alligator. A definitive symptom of structural pavement fatigue and failure in the underlying road base or subgrade.	**2.5 ×**
D40	Pothole	Bowl-shaped depressions or structural holes in the pavement surface resulting from moisture intrusion beneath cracked asphalt followed by freeze-thaw or traffic wear. Highly hazardous to traffic.	**3.0 ×**

3.3 Dual-Model Ensemble Architecture

Rather than relying upon a single general detector, Road Health AI pairs two complementary models:

The Dual-Model Ensemble:

- **Primary General Detector:** Fine-tuned on the full RDD2022 multi-national dataset to provide broad detection coverage across linear cracks (D00, D10) and complex alligator cracking patterns (D20).
- **Specialized Cavity & Pothole Model:** A higher-capacity model dedicated strictly to detecting deep road potholes, surface cavities, and edge rutting. This ensures zero missed potholes even under low-contrast or shaded conditions.
- **Intelligent Box Fusion:** Bounding boxes predicted by both models are merged through Non-Maximum Suppression (NMS), preserving the highest-confidence detections and removing duplicate predictions.

3.4 Asphalt Surface Validation Mechanism

A critical real-world requirement for automated inspection systems is rejecting invalid uploads. If an end-user uploads a picture of a car, a pet, an indoor document, or a blurred non-road image, the system's surface validator immediately intercepts the request. The request is rejected with a clear user-facing notice: *"The uploaded photo is not recognized as an asphalt road surface"*, preserving database purity and preventing erroneous repair alarms.

4. Mathematical Scoring Engine & Pavement Formulas

The platform computes an objective, mathematically transparent **Road Health Score** on a standardized scale from 0 to 100. This eliminates human subjectivity and gives municipal authorities an unambiguous index for budgeting and prioritization.

4.1 Component Mathematical Metrics

1. Damage Index (DI)

The Damage Index aggregates the structural impact of every detected defect. Each detected defect contributes to the index proportional to its severity weight multiplied by the artificial intelligence model's detection confidence:

Damage Index Equation:
Damage Index = Sum over all detections of [Class Weight × Confidence]

Example: An inspection with two potholes (Class Weight 3.0) detected at 83% and 20% confidence yields:
Damage Index = (3.0 × 0.83) + (3.0 × 0.20) = 2.49 + 0.60 = **3.09**

2. Affected Surface Coverage Percentage (Coverage %)

Measures the cumulative surface footprint of damaged asphalt relative to the total photographic frame area:

Surface Coverage Equation:
Coverage % = Minimum of 100.0 and [(Sum of All Bounding Box Areas / Total Image Area) × 100]

Example: If two pothole bounding boxes occupy 336,680 pixels on a 1600 × 1153 pixel image (1,844,800 total pixels):
Coverage % = (336,680 / 1,844,800) × 100 = **18.25%**

3. Overall Road Health Score (0 to 100 Scale)

The overall score starts at a pristine baseline of 100 points. Penalties are deducted based upon both the Damage Index and the Affected Surface Coverage Percentage:

Road Health Score Equation:
Road Health Score = 100.0 − (Damage Index × 3.0) − (Coverage % × 1.2)

Calculation for Inspection #109:
Score = 100.0 − (3.09 × 3.0) − (18.25 × 1.2)
Score = 100.0 − 9.27 − 21.90 = **68.83 / 100** (Condition: Fair)

4.2 Four Standard Condition Categories

Health Score Range	Condition Category	Visual Indicator	Recommended Engineering Action
81.0 – 100.0	**Good**	GOOD	Pavement is structurally sound. Zero or minor early-stage surface wear. Routine monitoring only.
61.0 – 80.0	**Fair**	FAIR	Moderate cracking or localized pothole. Seal coating, crack filling, and scheduled maintenance recommended.
31.0 – 60.0	**Poor**	POOR	Significant deterioration with multiple defects. Automated alert dispatched to local contractor for priority patching.
0.0 – 30.0	**Critical**	CRITICAL	Severe structural breakdown across multiple zones. Emergency repair protocol triggered; resurfacing needed.

5. Database Architecture & Complete Relational Schema

The persistent data storage layer is managed by an ACID-compliant relational SQLite database with three core tables.

5.1 Entity Overview & Relationships

- **Users to Analyses:** One-to-Many relationship. Each road scan is linked to the authenticated user account that submitted it.
- **Analyses to Email Logs:** One-to-Many relationship. Every email alert transmission attempt (whether to a contractor, admin, or citizen) records the referenced inspection ID.

5.2 Table: `users`

Field Name	Storage Type	Field Purpose & Business Rules
`id`	Integer (Primary Key)	Unique sequential identifier for each user account.
`email`	String (Unique, Indexed)	User's email address used for login and receiving inspection updates.
`password_hash`	String (Nullable)	Salted BCrypt cryptographic hash. Nullable for third-party OAuth logins.
`full_name`	String	Full human name or registered contractor corporate name.
`role`	String	Security permission level: `'user'`, `'contractor'`, or `'admin'`.
`is_active`	Boolean	Account operational status flag. Defaults to True.
`is_approved`	Boolean	Administrative verification flag. Automatically True for public users; defaults to False for contractors until approved by an administrator.
`service_state`	String (Nullable)	Contractor territorial state assignment (e.g., 'Maharashtra').
`service_city`	String (Nullable)	Contractor territorial city assignment (e.g., 'Pune').
`service_area`	String (Nullable)	Contractor neighborhood or sub-city ward jurisdiction (e.g., 'Swarget', 'Katraj').
`reset_otp_code`	String (Nullable)	Temporary 6-digit numeric one-time password for password reset verification.
`reset_otp_expires_at`	DateTime (Nullable)	Timestamp when the password reset OTP expires (10-minute validity window).
`created_at`	DateTime	Timestamp when the user account was registered.

5.3 Table: `analyses`

Field Name	Storage Type	Field Purpose & Business Rules
`id`	Integer (Primary Key)	Unique sequential assessment number (e.g., Inspection #109).
`user_id`	Integer (Foreign Key)	References `users.id` to enforce data ownership and history isolation.
`timestamp`	DateTime	UTC date and time when the analysis inference completed.
`latitude` / `longitude`	Float (Nullable)	GPS geographic coordinates pinned by user GPS or map selector.
`location_name`	String (Nullable)	Human-readable street address obtained via reverse geocoding.
`original_image_path`	String	Relative disk storage path to the raw uploaded road photograph.
`annotated_image_path`	String	Relative disk storage path to the AI-annotated photograph with bounding boxes.
`detections`	JSON / Structured	Array of all detected defects with class codes, names, confidence %, and pixel coordinates.
`damage_index`	Float	Calculated Damage Index metric for the pavement surface.
`coverage_pct`	Float	Calculated surface distress footprint percentage.
`severity`	String	Categorical rating: 'Low', 'Medium', or 'High'.
`road_health_score`	Float	Overall 0 to 100 Road Health Score.
`condition_label`	String	Condition tier: 'Good', 'Fair', 'Poor', or 'Critical'.

5.4 Table: `email_logs`

Field Name	Storage Type	Field Purpose & Business Rules
`id`	Integer (Primary Key)	Unique log entry record number.
`timestamp`	DateTime	Exact time when the SMTP email transmission was executed.
`recipient`	String	Target destination email address.
`subject`	String	Full email subject line dispatched through SMTP.
`status`	String	Transmission outcome flag: `'sent'` or `'failed'`.
`error_message`	String (Nullable)	SMTP diagnostic error string if transmission failed.
`analysis_id`	Integer (Nullable)	Inspection ID that triggered the email notification.

6. Detailed Breakdown of All Implemented Features & Tasks

The project encompasses sixteen major feature modules designed, implemented, and verified end-to-end:

1. Multi-Tier Role-Based Authentication & Authorization (RBAC)

- Three distinct security roles: **Public User**, **Contractor**, and **Administrator**.
- Protected API endpoints enforce authorization checks via secure JWT token headers.
- Administrator accounts are strictly protected: only designated administrative emails (such as `kauteshantanu80@gmail.com`) are permitted to register with the Admin role.

2. Contractor Registration & Administrative Approval Lifecycle

- Contractor sign-ups enter a `pending` status (`is_approved = False`) requiring admin verification.
- Administrators review pending contractors via a dedicated management table with one-click Approve, Reject, or Revoke controls.
- Contractors cannot access the contractor dashboard until approved by an administrator.

3. Contractor Territorial Jurisdiction Management

- Contractor profiles record three territorial fields: `service_state`, `service_city`, and `service_area`.
- Contractors can update their assigned coverage zones at any time through their dashboard.
- The system dynamically synchronizes approved contractor emails into the environment configuration to maintain continuous alert readiness.

4. Automated Pavement Distress Inference Pipeline

- Accepts high-resolution road photographs up to 25 MB with configurable confidence thresholds (0.15 to 0.60).
- Performs automated asphalt verification before tensor inference to reject non-road uploads.
- Generates visualizer overlays with color-coded bounding boxes, defect labels, confidence percentages, and condition headers.

5. Interactive Before/After Comparison Slider

- Interactive dual-handle slider on the results page allows inspectors to drag horizontally across the image to compare the raw road surface directly against the AI detection overlay.
- Supports three viewing tabs: **Before/After Slider**, **Annotated Bounding Boxes**, and **Original Road**.

6. Smart Damage Alert Triggering Engine

- Automatically triggers alerts whenever actionable road damage is detected:
  - Any road categorized as **Critical** or **Poor** (Score ≤ 60).
  - Any road categorized as **Fair** that contains detected defects (such as potholes or cracks, Score 60–80).
  - Any road with Score < 80 and detected defects.
- Clean, undamaged roads (Score ≥ 80, zero defects) do not dispatch alarm emails.

7. Hierarchical Geographical Contractor Routing

- Matches the inspection address against contractor jurisdictions in a 4-tier hierarchy: (1) Specific Area, (2) City, (3) State, (4) All Active Contractors.
- Ensures an alert for Swargate, Pune goes exclusively to the contractor responsible for Swargate, without sending alerts to contractors in Katraj or other areas.
- Incorporates transliteration mapping for Devanagari area names (e.g., matching *स्वारगेट* to *Swargate* / *Swarget*).

8. Administrator Direct Notification Pipeline

- Active administrators are automatically included on all damage alerts across all municipal regions.
- Ensures city administrators maintain real-time visibility into every structural road failure reported in their jurisdiction.

9. Citizen Inspection Submission Confirmation Emails

- Public users receive an automated confirmation email immediately upon submitting an inspection.
- The email details the Inspection ID, Road Health Score, and confirms whether a contractor has been notified for repairs.

10. Self-Service 3-Step Forgot Password Workflow

- Accessible across all roles (Admin, Contractor, User) via email verification:
  1. User requests a 6-digit numeric OTP sent directly to their email.
  2. User enters and verifies the OTP within a 10-minute security window.
  3. User sets a new password, receives an email confirmation, and logs in immediately.

11. Interactive GIS Road Damage Map

- Interactive Leaflet map displaying all geo-pinned road damage inspections across cities.
- Color-coded map markers represent pavement health: Green (Good), Yellow (Fair), Orange (Poor), Red (Critical).
- Clicking any marker opens an inspection summary popup with direct navigation to the full assessment report.

12. Contractor Priority Task Queue

- Contractor dashboard sorts road inspections by structural urgency: `Critical → Poor → Fair`.
- Displays a prominent "My Service Area" jurisdiction card highlighting the contractor's assigned region.

13. Administrator Governance & Analytics Dashboard

- System-wide analytics cards: Total Analyses, Average Health Score, Critical Roads Count, Active Contractors Count.
- Inspection management table, contractor approval control panel, and real-time email dispatch log viewer.

14. Real-Time Email Dispatch Audit Logging

- Every outgoing email transmission is logged in the `email_logs` table with recipient, subject, status, timestamp, and error details.
- Provides municipal administrators with a transparent audit trail of contractor notifications.

15. On-Demand "Notify Contractor" / Resend Alert Action

- Results page top bar features a **Notify Contractor** button.
- Allows administrators and inspectors to manually re-dispatch the exact assessment report to the assigned contractor on demand.

16. Live Camera Inspection with Automated Bottom-Stamping Geo-Tagging ("IO Tag Photo")

- **Live Camera Capture Stream:** Real-time in-browser video stream via WebRTC device media APIs with one-click front and rear camera toggling, plus native mobile camera hardware fallback.
- **Automated GPS Sensor Acquisition:** High-precision browser geolocation locks latitude and longitude down to 5 decimal places and automatically reverse geocodes the exact street address via OpenStreetMap Nominatim.
- **Permanent Canvas Geo-Tag Banner ("IO Tag"):** The system automatically stamps a high-contrast semi-transparent dark banner with a cyan accent stripe across the bottom of the photo canvas. The banner permanently burns Road Health AI identification, formatted GPS coordinates, street address, and live timestamp directly onto the image pixels.
- **Cross-Role Platform Accessibility:** Fully integrated and accessible across all three system roles:
  - **Public Citizens:** Instantly capture and report road hazards from anywhere on mobile or desktop without manual coordinate entry.
  - **Contractors:** Conduct on-site repair inspections and verify pavement fixes with tamper-resistant, GPS-stamped visual proof.
  - **Administrators:** Perform municipal infrastructure audits with immediate geographic authenticity.
- **Seamless Pipeline Integration:** The captured geo-tagged photo is processed directly by the dual-model YOLOv8 deep learning pipeline, populates the database records, and pins automatically onto the interactive GIS Road Map.

7. Contractor Area Routing & Automated Email Notification Engine

7.1 Multi-Tier Contractor Matching Logic

When a road inspection with distress is submitted, the email routing engine executes a 4-tier resolution sequence:

Contractor Resolution Sequence:

1. **Location Address Normalization:** The location address string is converted to lowercase. If Devanagari script is present, known English transliterations are appended (e.g., *स्वारगेट → swargate*, *कात्रज → katraj*).
2. **Tier 1 — Specific Area Match:** Active approved contractors are evaluated by their `service_area`. If a contractor's assigned area matches any token in the location address, the alert is sent exclusively to those area contractors. Unrelated area contractors are excluded.
3. **Tier 2 — City Match (Fallback):** If no contractor matches the specific area, contractors assigned to that `service_city` are selected.
4. **Tier 3 — State Match (Fallback):** If no contractor matches the city, contractors assigned to that `service_state` are selected.
5. **Tier 4 — Global Fallback:** If no territorial match is found, all active approved contractors receive the alert.
6. **Administrator Routing:** Active administrators are included on the recipient list, ensuring municipal leadership receives every damage notification.

7.2 Automated Test Shielding & Bounce Prevention

During automated test suite execution (such as CI/CD test scripts or integration runners):

- If an uploader or recipient email belongs to a test domain (such as `@test.com` or `@example.com`), the SMTP transmission is simulated.
- The dispatch is recorded in `email_logs` as `status = 'sent'` with the note *"Simulated dispatch for test domain"*.
- This prevents dummy test emails from attempting real SMTP delivery, eliminating Gmail bounce notifications and preventing test spam from reaching administrator inboxes.

8. Complete RESTful Application Programming Interface

(API) Catalog

HTTP Method & Route	Access Level	Route Purpose & Functionality
`GET /health`	Public	Health check endpoint returning backend status, system name, and version number.
`POST /auth/register`	Public	Registers a new account. Accepts email, password, full name, role, and contractor service area.
`POST /auth/login`	Public	Authenticates user credentials and returns a signed JSON Web Token (JWT Bearer token).
`GET /auth/me`	Bearer Token	Returns the authenticated user's profile, role permissions, and service jurisdiction.
`POST /auth/forgot-password/request-otp`	Public	Generates and emails a 6-digit numeric OTP code for self-service password reset.
`POST /auth/forgot-password/verify-otp`	Public	Validates the 6-digit OTP code against the database record within the 10-minute expiry window.
`POST /auth/forgot-password/reset`	Public	Updates the user's password after OTP verification and dispatches an email confirmation.
`POST /analyze`	Bearer Token	Uploads road photograph with optional GPS coordinates; runs YOLO inference and scoring.
`GET /history`	Bearer Token	Returns historical analyses. Citizens see only their own uploads; Contractors and Admins see all.
`GET /analyses/{id}`	Bearer Token	Fetches a single inspection assessment report and complete defect breakdown by ID.
`POST /analyses/{id}/resend-alert`	Bearer Token	Manually re-triggers the damage alert email to the assigned area contractor and administrator.
`GET /map/markers`	Public	Returns all geo-pinned road damage coordinates for public interactive map

HTTP Method & Route	Access Level	Route Purpose & Functionality
		display.
`GET /contractor/dashboard`	Contractor/Admin	Returns priority-sorted inspection queue for contractors (Critical → Poor → Fair).
`PUT /contractor/profile`	Contractor/Admin	Updates contractor service jurisdiction (service_state, service_city, service_area).
`GET /admin/dashboard`	Admin Only	Returns system-wide KPI metrics, health summaries, and active contractor statistics.
`GET /admin/contractors`	Admin Only	Lists all contractor accounts with approval status and service territory assignments.
`PUT /admin/contractors/{id}/approve`	Admin Only	Approves, rejects, or revokes contractor operational credentials.
`GET /admin/email-logs`	Admin Only	Audits all historical email dispatch records, delivery statuses, and transmission timestamps.

9. System Workflows, Deployment Guide & Verification

9.1 End-to-End User Inspection Workflow

Step-by-Step Inspection Lifecycle:

1. **Image Capture & Selection:** A citizen, contractor, or municipal inspector captures an asphalt road photograph using the live camera with automated GPS bottom-stamping or uploads an existing road photograph.
2. **Geographic Location Tagging:** The location is captured automatically via camera geo-tagging or selected via "Use My Location" (browser GPS) or "Pick on Map" to drop a pin. Nominatim reverse geocodes the coordinates into a street address.
3. **Asphalt Surface Validation:** The backend verifies that the image displays an asphalt road surface before proceeding.
4. **Deep Learning Inference:** The dual-model ensemble detects all surface distress features (D00, D10, D20, D40) with confidence percentages.
5. **Mathematical Metric Calculation:** The scoring engine calculates the Damage Index, Affected Coverage %, Severity Level, and 0–100 Road Health Score.
6. **Database Storage:** The assessment is saved with relational user ownership, timestamps, and bounding box coordinates.
7. **Automated Notifications:**
   - Citizen receives a submission confirmation email.
   - If damage is detected, the area-specific contractor and city administrator receive an urgent alert with the annotated image attached.
8. **Interactive Review:** The inspector explores the results page using the interactive Before/After comparison slider, defect breakdown table, and GIS damage map.

9.2 Verification & Test Results

The system includes comprehensive automated test suites that validate all platform modules:

- **Authentication & Security Suite:** Verifies user registration, contractor pending status, admin email protection, JWT issuance, and OTP verification.
- **Protected Endpoints Suite:** Verifies token-based access enforcement, analysis creation, and history isolation between user roles.
- **Geographic Mapping Suite:** Verifies coordinate persistence, map marker generation, and privacy preservation (zero user PII in public map markers).

- **Contractor Routing & Email Suite:** Verifies that damage alerts route to the correct territorial contractor and that test domain emails are safely simulated without bouncing.
- **Verification Outcome:** All 24 automated integration test checks pass with 100% green status. The frontend production bundle compiles cleanly in 32 seconds with zero errors.
